MUGUNTHAN R

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SKILLS

Mechanical Engineering

- **CAD & Design** - SolidWorks, 3D Modeling, 2D Drafting
- **Design Practices** - DFM, DFA, Tolerance Stack-up
- **Manufacturing** - CNC Machining, Precision Turning, Assembly
- **Connector Systems** - ARINC 801, MPO, Rugged Fiber Optics, MIL-DTL-38999
- **Enclosures** - Ruggedized Chassis, RFoF PCB Integration
- **Inspection** - CMM, Dimensional Validation, Reverse Engineering
- **Standards** - Defense & Aerospace, IP-Rated Designs

EXPERIENCE

WaveOptix Defence Solutions Pvt. Ltd. India
Mechanical Design Engineer *Sep 2023 - Present*

- Designed machined and fabricated rugged chassis and IP-rated enclosures for PCB systems
- Created SolidWorks 3D models, 2D drawings and BOMs for RFoF product packaging
- Performed CMM-based reverse engineering and dimensional validation
- Coordinated vendors for CNC machining, fabrication, assembly and inspection

Teraxis Evenings and weekends
Mechanical Design Engineer (Part-time) *Present*

- Developed ARINC 801 plugs and receptacles in 2, 4, 6, 8 and 12 channels on the first attempt
- Developed MPO plugs and square-flange receptacles in 1, 2, 3, 4, 6 and 8 channels
- Developed ruggedized 4-channel optical plugs, square-flange and jamnut receptacles
- Designed connector shells, inserts, alignment features and coupling interfaces in SolidWorks

Electrowing Technologies Pvt. Ltd. Bangalore, India
Mechanical Design Engineer Intern *Dec 2020 - Aug 2023*

- Developed SolidWorks 3D models and manufacturing drawings using DFM/DFA principles
- Prepared BOMs and supported CNC machining, fabrication and assembly
- Coordinated with vendors and supported testing activities

PROJECTS

Teraxis Part-time
Expanded Beam Connector and Optical Termini

- Expanded beam: CAD assembly modelled; not yet manufactured and planned as a future project
- Optical termini: current development of mechanical parts and the spring-loaded assembly

WaveOptix Defence Solutions Pvt. Ltd. Current
Ruggedized Chassis Design - RFoF PCBs

- Integrated PCB mounts, connector interfaces, sealing and grounding in CNC-machined enclosures
- Considered EMI/EMC requirements in chassis design, grounding and mechanical integration
- Supported manufacturing, assembly and inspection

EDUCATION

PA College of Engineering and Technology
Bachelor of Engineering - Mechanical Engineering *2020*

NV Polytechnic College
Diploma in Mechanical Engineering *2015*